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Idea Watch

The Science of Sensory Marketing
New research suggests that many industries are missing opportunities to connect with customers' senses. PLUS Trade is not as global as we think, why a dutiful spouse can boost your career, and more

probe

THE SCIENCE OF SENSORY MARKETING

New research suggests that many industries are missing opportunities to connect with customers' senses.

For two decades marketers in a variety of industries have been building expertise in reaching consumers through the five senses—learning to deploy cues, such as the sting from a swig of mouthwash and the scritch-scratch sound of a Sharpie pen, that can intensify perceptions of brands. The past year has brought a rush of interest in the subject among academics. New research suggests that we're about to enter an era in which many more consumer products companies will take advantage of sense-based marketing.

Much of the new research centers on "embodied cognition"—the idea that without our conscious awareness, our bodily sensations help determine the decisions we make. For example, people who had briefly held a warm beverage were more likely than people who had held a cold one to think that a stranger was friendly; this was demonstrated in an experiment by Lawrence E. Williams, of the University of Colorado at Boulder, and John A. Bargh, of Yale. And warm ambient temperatures prompted people to conform to a crowd, a finding of researchers led by Xun (Irene) Huang, of Sun Yat-sen University (see the exhibit below).

Marketing researchers are "starting to realize how powerful the responses to nonconscious stimuli can be," says S. Adam Brasel, an associate professor of marketing at Boston College. Work on embodied cognition has begun "blowing up on the academic side," he adds. At the 2014 Association for Consumer Research's North American conference, Brasel heard more papers on sensory research presented than at any previous conference. That same year the *Journal of Consumer Psychology* published a special issue on embodiment and sensory perception, with a focus on how sensory inputs can drive consumer behavior.

Aradhna Krishna directs the Sensory Marketing Laboratory at the University of Michigan and is considered the foremost expert in the field. She says that many companies are just starting to recognize how strongly the senses affect the deepest parts of our brains. The author of the 2013 book *Customer Sense: How the 5 Senses Influence Buying Behavior,* Krishna got into the field because she was fascinated by certain questions: Why does wine taste better in a wine glass than in a water glass? Why is an ad showing a piece of cake more engaging when the fork is placed to the right of the cake? Why does the smell of cinnamon make a heating pad seem to work better? Krishna realized that the senses amplify one another when they are congruent in some way. Because cinnamon suggests warmth, it can enhance a heating pad's appeal and apparent effectiveness. Such influences are subtle—and that's exactly why they are so powerful. Consumers don't perceive them as marketing messages and therefore don't react with the usual resistance to ads and other promotions.

Thinking about sensory effects is an established practice in some consumer industries, such as food, cosmetics, and hospitality. For example, Hershey's has long been aware that the tactile pleasure people get from unwrapping the foil around a Kiss transforms an ordinary piece of chocolate into a special experience. But many companies are taking their thinking much further. Consider this campaign by Dunkin' Donuts in South Korea: When a company jingle played on municipal buses, an atomizer released a coffee aroma. The campaign increased visits to Dunkin' Donuts outlets near bus stops by 16% and sales at those outlets by 29%. Another example is Olay Regenerist thermal facial products, which are engineered to generate heat upon application (although heat isn't necessary to their functioning) to signal that they are working.

Automakers have paid close attention to the senses for years: Designers work hard to optimize the feel of knobs, the solid noise of a door shutting, and the distinctive new-car smell. Recently they have turned to advanced technologies. For instance, in its 2014 M5 model, BMW mikes and amps the engine sounds through the car speakers, even when the audio system is turned off. The idea is to enhance the car's sporty feel.

WARMTH INDUCES PEOPLE TO CONFORM

Research participants in a warm room (versus a cool but not uncomfortable one) reported feeling closer to the people around them, and they were more likely to:

say they would buy the TV remote preferred by most of the other participants;

match their stock price predictions to those of previous participants;

and bet on the "favorite" in a hypothetical horse race.

SOURCE "WARMTH AND CONFORMITY: THE EFFECTS OF AMBIENT TEMPERATURE ON PRODUCT PREFERENCES AND FINANCIAL DECISIONS," BY XUN (IRENE) HUANG, MENG ZHANG, MICHAEL K. HUI, AND ROBERT S. WYER JR.

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Still, in wide swaths of consumer industries, companies remain focused solely on visual attributes and give little thought to other sensory effects. Product developers and marketers need to change that, Krishna says. Bank executives should make sure that branch offices exude the reassuring, wealth-suggesting aromas of wood and leather. Manufacturers of products with embedded motors should think about those products' sounds—are they tinny whines or solid, low-pitched hums? Luxury clothing manufacturers doing business online should consider what message is conveyed when goods are shipped in bubble wrap versus high-quality crinkly paper.

For managers looking to learn about sensory stimuli, the new academic work reveals striking instances of senses' affecting attitude, mood, and even memory more profoundly than words ever could. An experiment Krishna conducted with May O. Lwin, of Singapore's Nanyang Technological University, and Maureen Morrin, then of Rutgers University, is just one example. The three found that imbuing pencils with the unusual scent of tea tree oil dramatically increased research subjects' ability to remember the pencils' brand and other details. Whereas those given unscented pencils experienced a 73% decline in the information they could recall two weeks later, subjects given tea-tree-scented pencils experienced a decline of only 8%.

"In the past, communications with customers were essentially monologues—companies just talked at consumers," Krishna says. "Then they evolved into dialogues, with customers providing feedback. Now they're becoming multidimensional conversations, with products finding their own voices and consumers responding viscerally and subconsciously to them."

Such conversations, she emphasizes, should be at the center of product innovation and marketing for many brands. Every consumer company should be thinking about design in a holistic way, using the senses to help create and intensify brand personalities that consumers will cherish and remember.

THE IDEA IN PRACTICE

"TAKING SENSORY COMMUNICATION TO A WHOLE NEW LEVEL"

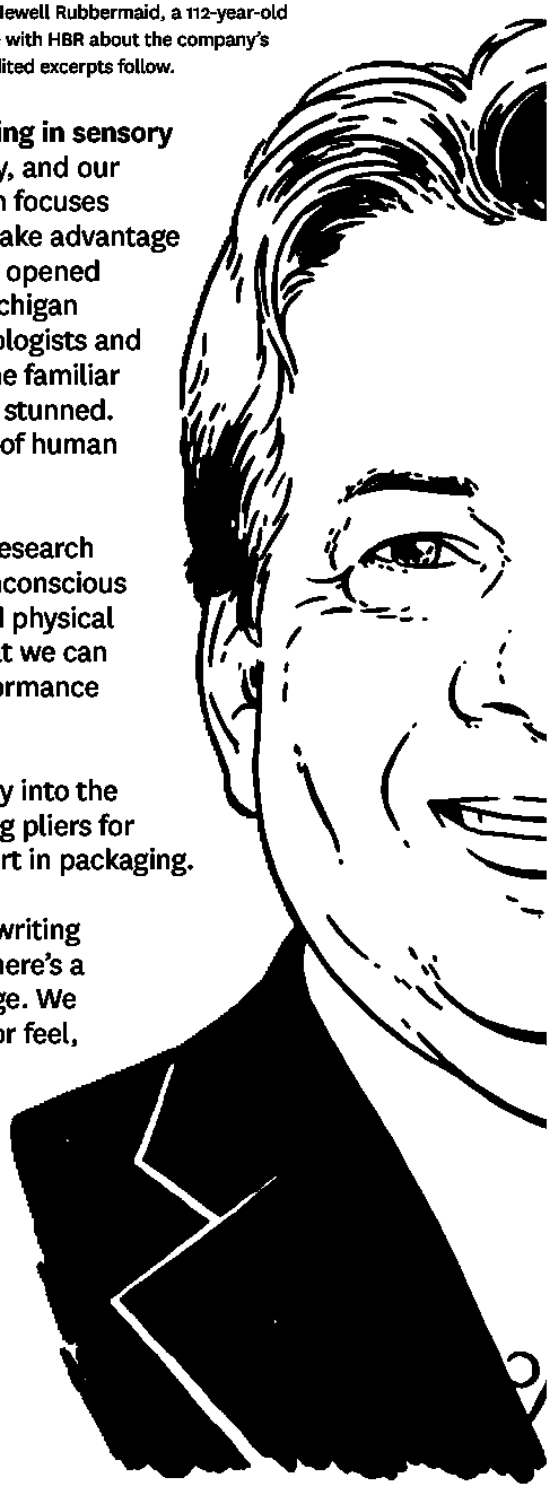
Chuck Jones is the chief design and R&D officer at Newell Rubbermaid, a 112-year-old maker of tools, pens, and other products. He spoke with HBR about the company's new emphasis on sensory marketing and design. Edited excerpts follow.

Why is Newell Rubbermaid investing in sensory inputs? The field is evolving quickly, and our corporate growth game plan, which focuses broadly on design, positions us to take advantage of the latest research. Last year we opened a design center next to Western Michigan University and staffed it with physiologists and also experts in perception. Someone familiar only with our past efforts would be stunned. And we've expanded our definition of human sciences research.

How so? Conventional marketing research looks at opinions. We also study unconscious behaviors and human cognitive and physical processing to extract principles that we can apply to Sharpie pens or high-performance pliers, for example.

Pliers? We're putting a lot of energy into the heft, feel, and ergonomics of cutting pliers for skilled trades. There's a similar effort in packaging.

Why packaging? Many of our fine writing instruments are given as gifts, so there's a ritual quality to opening the package. We pay close attention to the "hand," or feel, of the material; the resistance it presents (more is better, within limits); the sounds it makes; and the way the package opens, as a series of disclosures—an unveiling of the product, if you will. Our thinking about packaging takes sensory communication to a whole new level.



KYLE HILTON

TRADE NOT AS GLOBAL AS WE THINK

Our hyperconnected world isn't as tightly linked as it was during the peak of globalization, in 2007—just before the financial crisis hit. That's the message from the DHL Global Connectedness Index 2014.

The index uses flows of trade, capital, people, and information to show how entwined we citizens of the world are. It measures those flows along two dimensions: Depth reflects the volume of international activity, while breadth reflects its geographic distribution. For example, tourism in the Bahamas scores high on depth, because a lot of people travel there, but low on breadth, because most come from one country, the U.S.

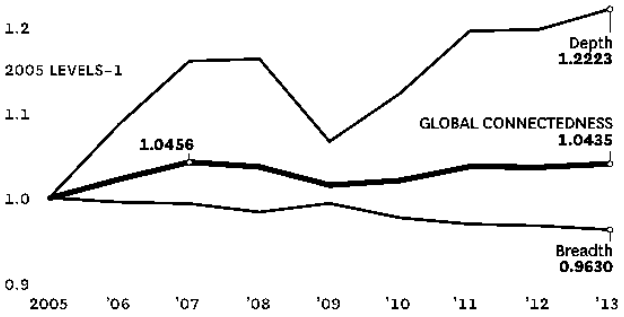
The index calculates the connectedness of each nation by combining depth and breadth. It also tracks connectedness on a worldwide level.

MANY LEADERS HAVE LOST GROUND

Fully half the 26 most-connected countries in 2013 have become less connected since 2007—and often the declines are steep. The financial crisis and recession caused trade flows to plummet. Capital has been largely flat, as has the number of people studying or working outside their home countries. Information flows have been rising fast, but they started from a low base; even now less than 20% of internet traffic crosses borders, and fewer than 5% of telephone calls do.

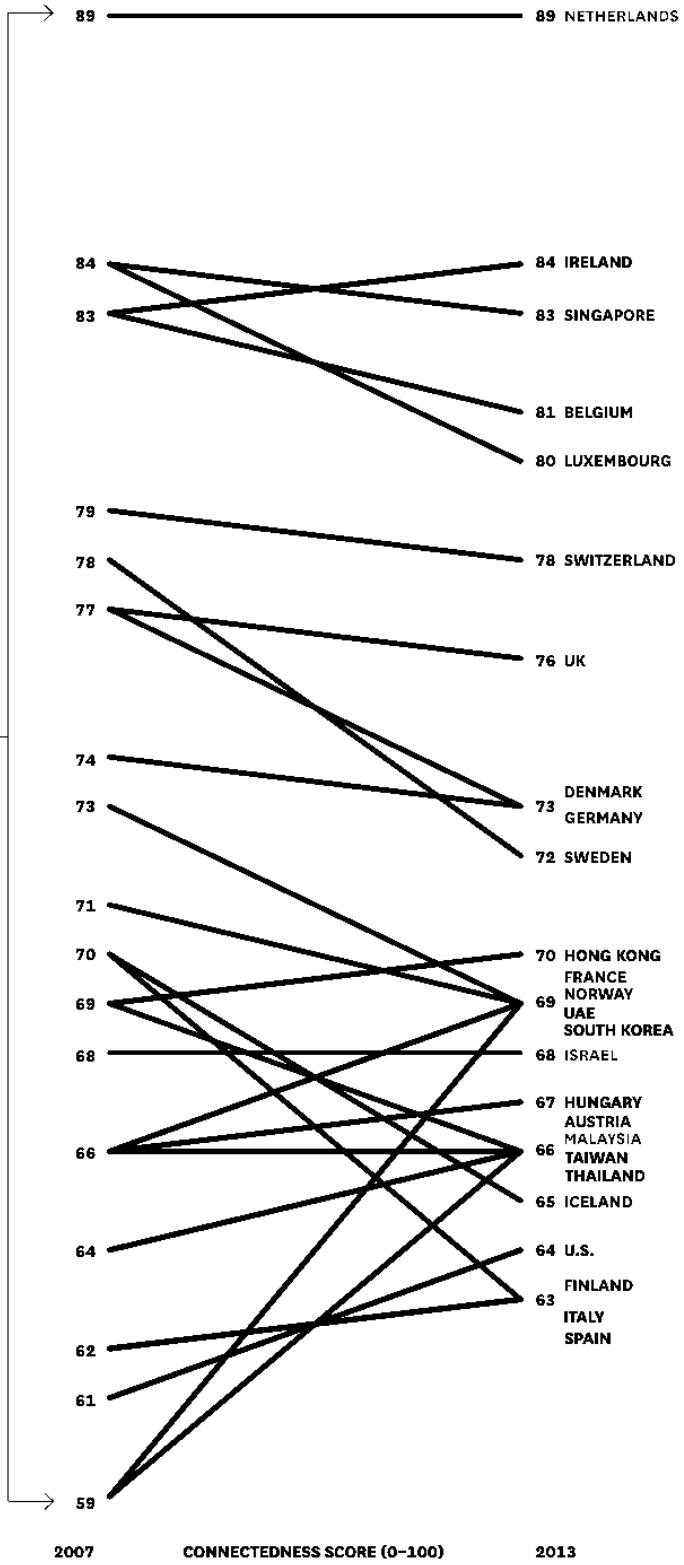
DEPTH BUT NOT BREADTH

Worldwide, the volume of flows has rebounded since the recession—but those gains have been offset by continued declines in geographic distribution.



SOURCE "DHL GLOBAL CONNECTEDNESS INDEX 2014," BY PANKAJ GHEMAWAT AND STEVEN A. ALTMAN

THE MOST-CONNECTED COUNTRIES

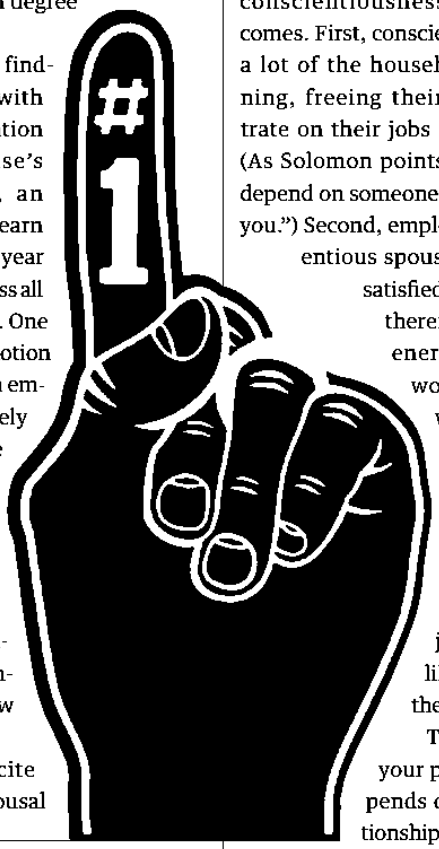


WORK & FAMILY
WHY A DUTIFUL SPOUSE CAN BOOST YOUR CAREER

For professionals trying to advance a career, a spouse can be a vital source of advice, commiseration, and support. But exactly what kind of spouse is most likely to help you get ahead? A new study by Brittany C. Solomon and Joshua J. Jackson, of Washington University in St. Louis, uses data collected on thousands of Australian households to analyze the effects of spouses' personality characteristics on people's employment outcomes.

The data covered what are known as the "big five" dimensions of personality—agreeableness, conscientiousness, extroversion, neuroticism, and openness. The researchers found that the only spousal trait that is important to an employee's work outcomes is conscientiousness, which predicts income, number of promotions, and job satisfaction, regardless of gender (and regardless of the employee's own degree of conscientiousness).

To put the income finding in dollar terms, with every standard-deviation increase in a spouse's conscientiousness, an employee is likely to earn about $4,000 more a year (the average figure across all ages and occupations). One way to frame the promotion finding is to say that an employee with an extremely conscientious spouse (two standard deviations above the mean) is 50% more likely to get a promotion than an employee with an extremely unconscientious spouse (two standard deviations below the mean).



The researchers cite three reasons why spousal conscientiousness boosts career outcomes. First, conscientious spouses handle a lot of the household chores and planning, freeing their partners to concentrate on their jobs or simply to recharge. (As Solomon points out, "When you can depend on someone, it takes pressure off of you.") Second, employees who have conscientious spouses typically feel more satisfied in their marriages and therefore have more mental energy to devote to their work. Third, employees with conscientious spouses tend to emulate those spouses' diligent habits, and the imitated characteristics seem to have a positive impact on both job satisfaction and the likelihood of promotion, the research shows.

This doesn't mean that your professional success depends on your being in a relationship. Plenty of single people shine at work, of course, and plenty of effective business leaders are unattached. In fact, research suggests that under certain circumstances, being single can help CEOs run their companies more effectively: Firms led by unmarried chief executives invest more aggressively than other firms and take greater risks overall.

Still, successful people often do turn out to have strong marriages. "When you're in a relationship, you're no longer just two individuals—you're this entity," Solomon says. The more solid the entity, the greater your advantage.

ABOUT THE RESEARCH "The Long Reach of One's Spouse: Spouses' Personality Influences Occupational Success," by Brittany C. Solomon and Joshua J. Jackson; "Marriage and Managers' Attitudes to Risk," by Nikolai L. Roussanov and Pavel G. Savor

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Some of these articles previously appeared in different form on HBR.org.

"Increasingly the business objectives of the manager will become unattainable unless the long-range planning includes, built right into it, the managing of men and the realization of basic ethical and spiritual values."

"INTEGRATION OF PEOPLE AND PLANNING," BY PETER F. DRUCKER (HBR, NOVEMBER–DECEMBER 1955)

THOMAS JAMES